ParseAnything — Implementation Specs (21 Tickets)

Companion to `read.md`; the two files are aligned (see the traceability table at the end). Implement **strictly in numeric order** — each ticket depends only on lower numbers.

Tiers: `MVP` = required for a demo-ready product · `ADV` = advanced features from `read.md` that add judge impact. Tickets 1–9, 11–14, 17, 20, 21 are MVP; tickets 10, 15, 16, 18, 19 are ADV.

Demo-ready (MVP) means: drop a mixed batch of files → live pipeline → click any output block to see its source bounding box → confidence heatmap → tables prove their own totals → charts and equations decoded → corrupt file fails cleanly in seconds → benchmark page shows measured results vs. baselines → runs from one `docker compose up`.

Locked Decisions (don't re-debate during the hackathon)

Area	Choice
Backend	Python 3.11, FastAPI, Pydantic v2, SQLAlchemy (SQLite → Postgres), `ProcessPoolExecutor` for page parallelism (Redis + Arq in ticket 19)
PDF	`pypdfium2` (render) + `pdfplumber` / `pdfminer.six` (text, words, tables) — permissive licenses; **no PyMuPDF** (AGPL)
OCR	RapidOCR (ONNX); Tesseract as fallback
Layout	Docling layout model if it installs cleanly in < 30 min, otherwise heuristic detector (ruled lines → table, big image → figure, math glyph density → equation)

Area	Choice
Table structure (scanned)	Table Transformer / Docling TableFormer; OpenCV line-detection fallback; VLM last resort
Math	pix2tex / Texify-class model; VLM fallback
VLM	Gemini free tier adapter + Ollama/Qwen2.5-VL adapter behind one interface; `none` mode degrades gracefully
Office/legacy/email	`python-docx`, `python-pptx`, `openpyxl`, stdlib `email`, `olefile`, LibreOffice headless
Frontend	Next.js + TypeScript + Tailwind + shadcn/ui, page-image/pdf.js canvas with SVG bbox overlay, Recharts, KaTeX, TanStack Query, Zustand
Deploy	Vercel (frontend) + Docker backend (HF Spaces/Render) + laptop + Cloudflare Tunnel fallback

Global conventions

- Bboxes are normalized `[x0,y0,x1,y1]`, 0..1, origin top-left.
- Every block: `id, type, page, reading_order, bbox, text|null, confidence, flags[], extractor, ladder_level`.
- Never invent text: confidence < 0.30 ⇒ `text=null`, flag `withheld`, attach crop.
- Ladder levels: L0 native → L1 CPU OCR/CV → L2 specialist model → L3 VLM. Escalate only when confidence is below threshold.
- Error envelope is always `{status:"error", error:{code,message,stage,recoverable,elapsed_ms}}`.
- Every new module ships with ≥ 1 unit test and a fixture under `backend/tests/fixtures/`.

Plan at a Glance

#	Ticket	Tier	Owner	Est (h)
1	Repo, Docker & env scaffolding	MVP	C	0.75
2	Schema, error codes & preflight validator	MVP	A	1.5
3	Orchestrator, API core, SSE & full persistence	MVP	C	3.5
4	PDF core: page analysis, native text, OCR, region routing	MVP	A	4
5	Reading order & hierarchy	MVP	A	2
6	Tables: extraction, header paths, cross-page stitching	MVP	B	4.5
7	VLM gateway + figures, charts & equations	MVP	B	4.5
8	Numeric integrity checker + self-heal	MVP	B	2.5
9	Format adapters: image, XLSX/CSV, DOCX/PPTX	MVP	C	3.5
10	Email (EML/MSG) with recursive attachments + legacy formats	ADV	B	2.25
11	Confidence, flags, telemetry & exports	MVP	A	3
12	Frontend shell, upload & live pipeline	MVP	D	2.5
13	Provenance Viewer + heatmap + flag panel	MVP	D	4
14	Table, chart, math viewers + cost meter	MVP	D	3
15	Ask-the-Document with bbox citations (+ optional voice)	ADV	B (+D UI)	3

#	Ticket	Tier	Owner	Est (h)
16	Review queue & human corrections	ADV	C (+D UI)	2
17	Eval corpus, metrics harness & Benchmark Arena	MVP	A + C + D	4
18	Python SDK, MCP server & API playground	ADV	A (+D UI)	2
19	Scale-out: Redis queue, Postgres, object storage, perf pass	ADV	C	1.5
20	Robustness tests & deployment	MVP	C	2.5
21	Full docs, pitch deck & demo rehearsal	MVP	D (+ all)	3.5
	Total			**≈ 60 h** (MVP tier ≈ 49 h)

Parallel lanes (4 people, 24 h):

- **A:** 2 → 4 → 5 → 11 → 17 (corpus/baselines) → 18
- **B:** (stub with fixtures until 4's regions exist) 6 → 7 → 8 → 10 → 15 (backend)
- **C:** 1 → 3 → 9 → 16 (backend) → 17 (harness) → 19 → 20
- **D:** 12 (mock API until 3 is up) → 13 → 14 → 15/16/18 UI → 21

Hour-12 checkpoint: tickets 1–6, 9, 11 working end-to-end on 5 documents, with 12–13 displaying them. **Freeze MVP scope here**, then take ADV tickets only as time allows.

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1 — Repo, Docker & env scaffolding

MVP · Owner C · 0.75h · Depends: —

Build

- Monorepo: `backend/`, `frontend/`, `sdk/`, `samples/`, `eval/`, `docs/`, `scripts/` (structure from `read.md` §23).

- `backend/pyproject.toml` (fastapi, uvicorn, pydantic, sqlalchemy, pypdfium2, pdfplumber, pdfminer.six, rapidocr-onnxruntime, opencv-python-headless, pillow, python-docx, python-pptx, openpyxl, olefile, rapidfuzz, apted, pytest, ruff).
- `backend/Dockerfile` (python:3.11-slim + LibreOffice headless + fonts + tesseract), `frontend/Dockerfile`, `docker-compose.yml` (api + frontend; `redis`, `postgres`, `minio`, `ollama` under optional profiles).
- `.env.example`: `VLM_PROVIDER=gemini|ollama|none`, `GEMINI_API_KEY`, `VLM_BUDGET_USD`, `VLM_PRICE_PER_CALL`, `GLOBAL_DEADLINE_S=55`, `MAX_PAGES`, `MAX_FILE_MB`, `DATABASE_URL`, `ELEVENLABS_API_KEY` (optional).
- CI: GitHub Action running `ruff` + `pytest`.

Done when

- [ ] `docker compose up` serves `GET /health` and the frontend root.
- [ ] `pytest` and `npm run build` pass on the empty skeleton; CI is green.

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2 — Schema, error codes & preflight validator

MVP · Owner A · 1.5h · Depends: 1

Build

- Pydantic models: `Document, Page, Block, TableBlock (cells with row/col/rowspan/colspan/is_header/col_path/row_path/bbox/confidence, checks[], spans_pages), FigureBlock (kind, chart_type, series, axes, replot_agreement, description), EquationBlock (latex), ErrorReport, ParseResult` (with `parent_id`/children for email attachments).
- Enums for block types, flags (`low_confidence, needs_review, withheld, ocr, merged_cross_page, healed, human_verified`), and error codes: `E_UNSUPPORTED_FORMAT, E_CORRUPT_FILE, E_ENCRYPTED, E_EMPTY_DOCUMENT, E_TOO_LARGE, E_TIMEOUT_PARTIAL, E_OCR_UNAVAILABLE, E_VLM_UNAVAILABLE, E_INTERNAL`.
- Export `docs/schema.json` (`schema_version="1.0"`).
- Preflight (< 2 s): magic-byte sniffing (PDF, PNG, JPG, TIFF, OOXML zip, legacy OLE, EML, CSV heuristic, HTML), extension mismatch, zero-byte,

size/page limits, encrypted PDF, open-test with the right library.

Done when

- [ ] Fixtures for corrupt PDF, encrypted PDF, 0-byte file, and a renamed `.exe → .pdf` each return the correct error code in under 2 s (timing asserted).
- [ ] A sample `ParseResult` round-trips and validates against `schema.json`.

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3 — Orchestrator, API core, SSE & full persistence

MVP · Owner C · 3.5h · Depends: 2

Build

- Orchestrator with stages `preflight → detect → extract → assemble → verify → score → export`; page-level parallelism; async event bus; stage handlers are pluggable stubs so tickets 4–11 drop in without touching it.
- Global deadline `GLOBAL_DEADLINE_S` (55 s): on expiry return `status="partial"` with a per-page completion map.
- Endpoints: `POST /v1/parse` (async, plus `?sync=true`), `GET /v1/jobs/{id}`, `GET /v1/jobs/{id}/events` (SSE), `GET /v1/documents/{id}`, `GET /v1/health` (models loaded, VLM reachability).
- SSE events: `stage`, `page_classified`, `region_routed` (type, bbox, `route_reason`), `region_done` (confidence, ladder level), `check_result`, `done|error`.
- **Full DB schema from `read.md` §7** via SQLAlchemy models: `documents` (with `parent_id`), `jobs`, `pages`, `blocks`, `tables`, `table_cells`, `figures`, `equations`, `reviews`, `benchmark_runs`; indexes on `sha256` and `(document_id, reading_order)`. SQLite default, Postgres via `DATABASE_URL`.
- Page and crop images stored on disk behind a storage interface (S3-compatible swap in ticket 19); sha256 result cache (same file ⇒ instant result).

Done when

- [ ] Uploading a file returns a `job_id` immediately; SSE streams stage events in order.
- [ ] An artificially slow page triggers a `partial` result at the deadline.
- [ ] Re-uploading an identical file returns the cached result in < 1 s; restart preserves documents.

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4 — PDF core: page analysis, native text, OCR, region routing

MVP · Owner A · 4h · Depends: 3

Build

- Rasterize pages with `pypdfium2` (configurable DPI).
- Page analyzer: text-layer coverage, image-coverage ratio, skew estimate, blur/contrast quality ⇒ classify `digital | scanned | hybrid`.
- Native text: words/lines with font size, weight and bbox via `pdfplumber`; fix hyphenation and ligatures.
- OCR (RapidOCR) with deskew/denoise, word boxes and confidences; line/paragraph grouping.
- Region detector + router: type each region (`text, heading, list, table, figure, equation, header, footer`) and attach a human-readable `route_reason` (e.g. "image-only page → OCR"). Heuristic fallback if the layout model isn't available.
- Hybrid pages keep native text and route embedded images through figure/OCR paths.

Done when

- [ ] Digital, scanned and hybrid fixtures are classified correctly.
- [ ] A scanned fixture yields non-empty text with word-level confidences; an 8°-skewed image is still readable.
- [ ] Every region carries `type`, `bbox` and `route_reason`.

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5 — Reading order & hierarchy

MVP · Owner A · 2h · Depends: 4

Build

- Running header/footer detection by cross-page repetition (≥ 50% of pages, similar y) ⇒ tagged, excluded from the body flow.
- Column detection (x-range clustering + whitespace gutters) with recursive XY-cut; footnotes (bottom, small font) linked to markers; sidebars (narrow, offset) emitted after the main column they sit beside.
- Global integer `reading_order` across pages.
- Heading levels from font-size/weight clustering + numbering patterns; nested lists; caption ↔ figure/table linking by proximity and "Figure/Table N".

Done when

- [ ] Two-column and sidebar fixtures produce the expected order (normalized edit distance ≤ 0.1 vs. a hand-written gold sequence).
- [ ] Running headers and page numbers never appear in the body flow.
- [ ] Heading levels and caption links are correct on the sample paper.

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6 — Tables: extraction, header paths, cross-page stitching

MVP · Owner B · 4.5h · Depends: 4

Build

- **Digital tables:** `pdfplumber` lattice (ruled) and stream (whitespace) strategies; cell grid with `rowspan/colspan`, per-cell bbox and text.
- **Scanned tables:** Table Transformer / TableFormer on the rasterized crop for rows/columns/spans, map OCR words into cells; OpenCV line-detection + alignment clustering as fallback; VLM fallback (ticket 7) when structure confidence is low — leave a hook now.
- Header detection (bold/shaded/numeric-ratio change) and `col_path` / `row_path` resolution for multi-row headers (e.g. `["FY2024","Q1"]`).

- **Cross-page stitcher:** merge when the table ends near the page bottom (~8%), the next starts near the top (~8%), column counts match and x-positions agree within ±3%; drop the repeated header; keep per-page bboxes; set `spans_pages` and flag `merged_cross_page`; never merge when the evidence is weak.
- Outputs: HTML (always), Markdown (falls back to HTML when spans exist), CSV.

Done when

- [ ] Merged-cell fixture reproduces spans; two-level header fixture gives the right `col_path` for every data cell.
- [ ] A 3-page table becomes **one** table with the correct row count and no duplicate header; a negative fixture (two different tables) is **not** merged.
- [ ] Scanned-table fixture reaches ≥ 90% cell-text match on the gold sample.

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7 — VLM gateway + figures, charts & equations

MVP · Owner B · 4.5h · Depends: 4

Build

- **Gateway:** one interface, adapters for Gemini and Ollama, plus `none` (returns `E_VLM_UNAVAILABLE` metadata, pipeline continues). JSON-schema-constrained prompts, retries with backoff, sha256(crop+prompt) cache, per-job budget guard, cost accounting (price from env, never hardcoded).
- **Figures:** crop figure regions, classify (`chart|diagram|photo|logo`), VLM description, link caption.
- **Chart digitizer:** type detection → OCR axis ticks/labels/legend → pixel→value fit (report R^2) → bar contours / line color-path sampling / pie angles → prefer printed data labels → VLM structured-JSON cross-check → agreement score. Output `series[{name, points[{x,y}]}]`, `axes`, `chart_type`.
- **Re-plot verification:** redraw the extracted series in the original pixel frame and compute `replot_agreement`; save the overlay image.

- **Equations:** LaTeX-OCR model with VLM fallback; validate that the LaTeX parses (KaTeX-compatible), otherwise flag instead of emitting.
- PPTX/DOCX native chart data (ticket 9) bypasses the digitizer when present.

Done when

- [ ] Switching `VLM_PROVIDER` needs no code change; cache hit avoids a second call; budget exhaustion returns a clean result and the pipeline continues.
- [ ] On 5 sample charts (bar, grouped bar, line, pie + one more), mean relative error ≤ 5% where data labels exist; series count correct on ≥ 4/5.
- [ ] `replot_agreement` ≥ 0.9 on clean charts and visibly lower for a deliberately wrong series.
- [ ] 10-equation fixture ≥ 70% normalized match; invalid LaTeX is flagged, never silently emitted.

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8 — Numeric integrity checker + self-heal

MVP · Owner B · 2.5h · Depends: 6, 7

Build

- Number normalizer: parentheses negatives, currency (`$ ₹ € £`), `%`, thousands separators, units from header/caption ("in millions").
- Detect total/subtotal rows and columns; recompute with rounding tolerance; emit `checks[] {type,row|col,expected,found,ok}`.
- Mismatch ⇒ lower table confidence and identify suspect cells (lowest cell confidence in that row/column).
- **Self-heal:** crop only the suspect cells, re-read via the VLM with a strict "digits only" schema, re-run the check; accept the change **only** if it fixes the check; flag `healed` and keep both values; cap VLM calls per table.

Done when

- [ ] Correct-totals fixture: all checks ✓. One-digit-altered fixture: exactly one ✗ pointing at the right cell.
- [ ] Seeded OCR-error fixture goes ✗ → ✓ automatically; no change is applied when the VLM output doesn't fix the check.

9 — Format adapters: image, XLSX/CSV, DOCX/PPTX

MVP · Owner C · 3.5h · Depends: 4, 7

Build

- **Images** (PNG/JPG/multi-page TIFF): EXIF rotation, then the normal page analyzer/OCR/layout path.
- **XLSX/CSV/TSV:** each sheet ⇒ table block(s); merged cells ⇒ spans; header detection; cached formula values with the formula string in `meta`; provenance is `sheet` + `cell_range` (e.g. `A1:F20`); CSV delimiter auto-detected.
- **DOCX/PPTX dual path:** native structure via `python-docx` / `python-pptx` (headings, lists, tables, images, speaker notes, embedded chart data) **plus** LibreOffice render to PDF; align native blocks to rendered blocks (fuzzy text match + order) to recover `page` + `bbox`.
- Unsupported or conversion-failed inputs return the standard error envelope.

Done when

- [ ] DOCX and PPTX fixtures yield blocks with page + bbox; alignment succeeds for ≥ 90% of blocks; PPTX native chart data matches the source.
- [ ] Multi-sheet XLSX fixture emits one table per sheet with correct merged spans and cell ranges.
- [ ] A photographed printed table produces a table block through the image path.

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10 — Email (EML/MSG) with recursive attachments + legacy formats

ADV · Owner B · 2.25h · Depends: 9

Build

- **Email:** parse headers (from/to/date/subject), plain/HTML body, inline images; each attachment parsed as a **child document** through the same entry point (`parent_id` set); depth limit 3; zip-bomb and size guards. EML via stdlib `email`; MSG via `olefile`-based extraction.
- **Legacy converter:** `soffice --headless --convert-to` for DOC / PPT / XLS / RTF / ODT → modern format, then reuse ticket 9 adapters; timeout-guarded.
- Conversion failure ⇒ `E_CORRUPT_FILE` with stage `detect`.

Done when

- [ ] An EML with a PDF and an XLSX attachment outputs body blocks plus two child documents; JSON shows parent/child links.
- [ ] `.doc`, `.xls`, `.ppt` fixtures parse; a corrupted legacy file returns a structured error.

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11 — Confidence, flags, telemetry & exports

MVP · Owner A · 3h · Depends: 5, 6, 7, 8, 9

Build

- Confidence: `conf = 0.45·extractor + 0.25·agreement + 0.20·structural + 0.10·quality` (weights in config). Agreement = text-layer-vs-OCR sample on digital pages, two-extractor agreement for tables/charts; structural = sum-checks, axis-fit R^2, column consistency.
- Policy: `≥0.80 ok` · `0.50–0.80 low_confidence` · `<0.50 needs_review` · `<0.30 withheld` (text=null + crop). Thresholds configurable.
- Telemetry per region: `ladder_level, extractor, elapsed_ms, est_cost_usd`; per document: level histogram, escalation rate, pages/sec, total cost.
- **Markdown export:** reading order; headings, lists, GFM-or-HTML tables, figures with extracted data table, equations as `$$…$$`; provenance anchors `<!-- b:b_0008 p:3 bbox:… -->`; flagged blocks render as `> ⚠ [needs review — p3]`, never as invented text.
- **JSON export** validated against `schema.json` before return (invalid state ⇒ `E_INTERNAL`); includes child documents for emails.

- Endpoints: `GET /v1/documents/{id}/markdown`, `/pages/{n}/image`, `/blocks/{block_id}` (+ crop), `/tables/{block_id}?format=csv|html|json`, `/flags`, `GET /v1/cost/estimate`.

Done when

- [ ] A blurred fixture produces `needs_review` / `withheld` blocks; a clean digital fixture has ≥ 95% of blocks ≥ 0.80.
- [ ] Markdown anchors map 1:1 to JSON block ids; all fixtures validate against the schema.
- [ ] `/jobs/{id}` exposes live cost and the ladder-level histogram.

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12 — Frontend shell, upload & live pipeline

MVP · Owner D · 2.5h · Depends: 3 (develop against mocked events until then)

Build

- Next.js app, dark "forensic lab" theme (`#0B0F14`, cyan accent, green/amber/red confidence palette), shadcn/ui, Inter + JetBrains Mono, top bar, Framer Motion defaults.
- Route stubs for `/`, `/doc/[id]`, `/arena`, `/review`, `/docs`.
- Landing page: multi-file drag-drop + **"Try the Torture Test"** one-click sample set.
- Per-file live pipeline tiles consuming SSE: stage chips, page tiles, regions drawing themselves as `region_routed` events arrive; error state shows code + message.
- API client, SSE hook with reconnect, global store.

Done when

- [ ] Dropping 6 files shows 6 live pipelines; a corrupt file shows its error within seconds; partial results never crash the UI.

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13 — Provenance Viewer + heatmap + flag panel

MVP · Owner D · 4h · Depends: 11, 12

Build

- Three panes: page thumbnails with confidence dots · page image/PDF canvas with SVG bbox overlay · output tabs (Markdown / JSON / Tables / Charts / Math / Flags).
- **Bi-directional sync:** click a block in the output ⇒ scroll + highlight its bbox; click a bbox ⇒ scroll the output to the block.
- Hover tooltip: type, confidence, extractor, ladder level, `route_reason` ("Smart Router Explainer").
- Cross-page tables highlight on each page they span; spreadsheets show a `sheet!A1:F20` chip instead of a bbox; email documents show parent/child navigation.
- **Confidence heatmap** toggle (green/amber/red overlay); withheld blocks show hatched overlay + lock icon ("Refuses to guess").
- **Flags tab:** list of `needs_review` / `withheld` blocks with crop and a "mark verified" action (local state until ticket 16 wires it to the API).

Done when

- [ ] Sync works both ways on every demo fixture; overlay stays aligned at all zoom levels.
- [ ] Clicking a flag jumps to its region; heatmap toggles instantly; a 30-page doc scrolls smoothly.

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14 — Table, chart, math viewers + cost meter

MVP · Owner D · 3h · Depends: 13

Build

- **Tables:** spans rendered, header paths on hover, cell click ⇒ source crop, sum-check badges (✓ / ✗ / healed) with the ✗ → healing → ✓ animation, CSV/HTML download.
- **Charts:** extracted data table + Recharts re-plot; **original ⇄ overlay slider** driven by the stored overlay image and `replot_agreement`.
- **Math:** KaTeX render beside the source crop; invalid LaTeX shows a warning.

- **Cost meter:** odometer-style live `$` and an L0–L3 stacked bar of regions by ladder level (from telemetry); "why routed" popover.

Done when

- [ ] Each viewer works on the torture-test fixtures; downloads are valid files.
- [ ] Cost/level numbers match `/jobs/{id}` final values.

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15 — Ask-the-Document with bbox citations (+ optional voice)

ADV · Owner B (backend) + D (UI) · 3h · Depends: 11, 13

Build

- `POST /v1/documents/{id}/ask`: retrieve relevant blocks (BM25, optional embeddings), answer via the VLM/LLM constrained to cite `block_id`s; refuse with "not found in document" and **no citations** when evidence is insufficient.
- Chat panel in the viewer; citation chips jump to and highlight the source bbox.
- **Optional voice (feature flag, only if time):** ElevenLabs TTS play button on answers; check current free-tier limits first.

Done when

- [ ] Answers include ≥ 1 `block_id` citation and a citation click always lands on the correct bbox.
- [ ] An unanswerable question returns the "not found" state with no citations; with the flag off, no voice UI appears.

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16 — Review queue & human corrections

ADV · Owner C (backend) + D (UI) · 2h · Depends: 13

Build

- `POST /v1/blocks/{id}/review` stores original/corrected text in `reviews`; the block gets flag `human_verified` and its text is updated; `GET /v1/documents/{id}/flags` lists pending items.
- `/review` page: queue of flagged blocks across documents with crop, editable text, approve/correct actions; the Flags tab in the viewer now calls the API.
- Export of corrections as eval data (`eval/corrections.jsonl`) so they can feed the gold set.

Done when

- [ ] A correction persists across restart and updates the viewer, Markdown and JSON.
- [ ] Corrections appear in `corrections.jsonl` with source document, block id, original and corrected text.

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17 — Eval corpus, metrics harness & Benchmark Arena

MVP · Owner A (corpus, baselines) + C (harness, API) + D (Arena page) · 4h · Depends: 11 (and 10, 16 optional)

Build

- **Corpus:** ~20–30 documents from public/permissive sources (SEC filings, arXiv papers, scans, Office files); lightweight gold: table structures (HTML), a few block boxes/orders, chart data points, LaTeX strings; manifest with source/license. Optionally add an OmniDocBench subset (check license).
- **Torture set (6 files) in `samples/torture_test/`:** skewed scanned debt schedule, two-column paper with equations, multi-page table with repeated headers, PPTX with chart, **EML with a PDF attachment** (if ticket 10 is done, otherwise an XLSX with merged cells), corrupt PDF.
- **Harness** (`python -m eval.runner --corpus gold --system parseanything`): text NED/CER, reading-order NED, TEDS (+ struct), cell F1, bbox IoU@0.5/0.75, chart relative error, p50/p95 latency, pages/sec, escalation rate, $/1k pages (formula from telemetry), confidence reliability bins/ECE.

- **Baselines:** `pypdf` plain text and `pdfplumber` default (no bboxes ⇒ 0 on provenance).
- **Arena:** `POST /v1/benchmark/run`, `GET /v1/benchmark/runs/{id}` (stored in `benchmark_runs`); `/arena` page with side-by-side metrics, bar charts and reliability plot; **pre-computed run loads instantly**, "run live" works on a small corpus.

Done when

- [ ] Harness runs end-to-end; metric functions are unit-tested against hand-computed examples.
- [ ] Arena compares ≥ 3 systems with identical metric code; displayed numbers equal harness output.
- [ ] Measured escalation rate and $/1k pages are recorded for the pitch (**no unmeasured numbers on slides**).

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18 — Python SDK, MCP server & API playground

ADV · Owner A (SDK, MCP) + D (playground UI) · 2h · Depends: 11

Build

- `sdk/python`: `parseanything.parse("file.pdf")` → typed result with blocks, tables, provenance helpers.
- `sdk/mcp_server`: tools `parse_document`, `get_block`, `ask_document` (the last only if ticket 15 is done); returns blocks with page/bbox provenance.
- `/docs` page: API playground (upload a file, view request/response, copy `curl`/SDK snippets) linking to the FastAPI OpenAPI docs.

Done when

- [ ] The 3-line SDK example in the README runs; an MCP tool call returns blocks with provenance.
- [ ] Playground parses a sample and shows the raw JSON response.

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19 — Scale-out: Redis queue, Postgres, object storage, perf pass

ADV · Owner C · 1.5h · Depends: 17

Build

- Optional Redis + Arq worker mode (`QUEUE_BACKEND=redis`) alongside the default in-process pool; Postgres via `DATABASE_URL` (e.g. Supabase free tier); S3-compatible storage (MinIO) behind the storage interface.
- Performance pass using the harness: DPI, process-pool size, lazy model loading, cache hit paths; record p50/p95 and pages/sec with hardware noted in `docs/COST_PERF.md`.

Done when

- [ ] `docker compose --profile scale up` runs the API with Redis workers, Postgres and MinIO and passes the same smoke test.
- [ ] No accuracy regression in the harness after tuning; performance numbers are recorded.

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20 — Robustness tests & deployment

MVP · Owner C · 2.5h · Depends: 17 (and any ADV tickets already built)

Build

- Failure-mode tests (asserting < 60 s each): corrupt, truncated, encrypted, empty, huge page count, mismatched extension, zip bomb, VLM down, OCR missing, deadline expiry (plus corrupt attachment and legacy-conversion failure if ticket 10 exists).
- Deploy: frontend → Vercel; backend Docker (models baked into the image) → HF Spaces/Render; CORS verified.
- **Demo insurance:** full stack on the team laptop + Cloudflare Tunnel; `scripts/seed_demo.py` pre-parses the torture set; verify `VLM_PROVIDER=none` and offline (`ollama`) modes degrade gracefully.

Done when

- [ ] All failure-mode tests pass in CI with standard error/partial responses.

- [ ] Public URL parses a sample end-to-end; cold start < 60 s; offline `docker compose up` demo works.

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21 — Full docs, pitch deck & demo rehearsal

MVP · Owner D (+ all) · 3.5h · Depends: 20

Build

- `README.md` per `read.md` §21 (what/why, features, quick start with `docker compose up` + one `curl`, formats, output sample, pipeline + ladder diagram, benchmarks, cost/perf, config table, SDK/MCP usage if built, structure, limitations, licenses).
- `docs/` per `read.md` §20: `ARCHITECTURE`, `SCHEMA`, `API`, `ROUTING`, `CONFIDENCE`, `TABLES`, `CHARTS_MATH`, `BENCHMARKS`, `COST_PERF`, `SELF_HOSTING`, `LICENSES` (every dependency + license; confirm no AGPL/GPL in the core path), `LIMITATIONS`. Fill numbers only from measured results.
- 12-slide deck (`read.md` §19), 5-minute demo script with timings (§16), 2-minute pitch (§17), Q&A cheat-sheet (§18) updated to reflect which ADV features actually shipped; screen-recorded backup video.
- Code freeze at hour 22, then 3 rehearsals: deployed URL, laptop fallback, Wi-Fi off.

Done when

- [ ] A new dev runs the project from the README in < 10 min.
- [ ] Three clean rehearsals; backup video plays offline; release tag `v1.0-hackathon` pushed.

Cut-Line (if time runs short)

Drop in this order (matches `read.md` §15):

1. Voice/TTS (ticket 15, flag off)
2. SDK + MCP server (ticket 18; keep the playground only if it's cheap)
3. Ask-the-Document (ticket 15)
4. Email attachments, then legacy formats (ticket 10)

5. Redis/Postgres/MinIO scale-out (ticket 19; keep the perf pass)
6. Scanned-table model — use OpenCV + VLM instead (ticket 6)
7. Self-heal, keep the ✓/✗ checks (ticket 8)
8. Review queue persistence — keep the local-state Flags tab (ticket 16)

Never cut: Provenance Viewer (13), confidence/flags (11), fail-safe errors (2, 20), cross-page tables (6), benchmark numbers (17).

Traceability: `read.md` → tickets

`read.md` item	Ticket(s)
§3 #1 Click-to-verify Provenance Viewer	13
§3 #2 Self-verifying tables / self-heal	8, 14
§3 #3 Chart re-plot verification	7, 14
§3 #4 Cost-aware ladder + cost meter	4, 11, 14
§3 #5 Dual-path Office parsing	9
§3 #6 Email with recursive attachments	10
§3 #7 Cross-page table stitching	6
§3 #8 Benchmark Arena	17
§3 #9 License-clean self-hostable stack	1, 20, 21
§3 #10 Review queue	16
§3 #11 MCP server + SDK	18
§4 WOW: chart→data, equation→LaTeX	7
§4 WOW: confidence heatmap, "refuses to guess"	11, 13
§4 WOW: Ask-the-Doc, voice	15
§4 WOW: router explainer, semantic hierarchy	4, 5, 13

`read.md` item	Ticket(s)
§6.1–6.3 detection, reading order, stitching	4, 5, 6
§6.4 integrity checker	8
§6.5 chart digitizer	7
§6.6 confidence model	11
§6.7 fail-safe & error codes	2, 3, 20
§6.8 format matrix (PDF, images, Office, legacy, spreadsheets, email, HTML)	4, 9, 10
§7 database schema	3 (+ 16 reviews, 17 benchmark_runs)
§8 API endpoints	3, 11, 15, 16, 17
§9–10 frontend pages and UX	12, 13, 14, 15, 16, 17, 18
§12 advanced: Redis/Postgres, offline mode, caching	3, 19, 20
§14 deployment + demo insurance	20
§15 roadmap, cut-line	Plan at a Glance, Cut-Line
§16–18 demo, pitch, Q&A	21
§19–23 PPT, tech docs, README, resume, folder structure	1, 21
§24 evaluation methodology	17

Format note: HTML/TXT/MD ingestion from `read.md` §6.8 is a small addition to ticket 9's adapters (DOM → typed blocks, provenance = DOM path) if time allows; it is not a standalone ticket.